PRACTICAL 15

Final Integration, Testing, Documentation, GitHub Submission, and Local Deployment

AIM

To integrate StudentHub modules, document the local deployment, provide repeatable integration checks, and prepare the repository for a final review.

Introduction

The portal includes authentication, student CRUD, event management, a JSON API and an admin dashboard with audit logging. Practical 15 adds a deployment guide and smoke-check script. The local XAMPP site and configured app database connection are currently available for public and unauthenticated checks.

Tools and Technology

VS Code, XAMPP, PHP, MySQLi, MariaDB, HTML, CSS, JavaScript Fetch API, browser, Git and GitHub.

Files Used

File	Purpose
docs/Practical15-deployment.md	Local setup, integration checklist and Apache virtual-host notes.
scripts/check-practical15.py	Public-page and admin-demo access smoke checks.
docs/Practical15-test-results.json	Records whether local smoke checks ran.
docs/Practical13-setup.md	API setup and test guidance.
docs/Practical14-setup.md	Dashboard audit migration and setup steps.
README.md	Project index updated for Practicals 13-15.

Key Skills Used

Full-stack integration, test planning, deployment documentation, accessibility review, local Apache configuration and repository hygiene.

Implementation Overview

1 Integrated Modules

Existing authentication and role checks connect to student management, events and registrations. The Practical 13 API integrates event CRUD, and Practical 14 adds dashboard statistics and audit activity.

2 Repeatable Verification

The API test script uses a temporary event and removes it after testing. The local smoke script checks public pages and unauthenticated access to the admin demo. Existing Practical 10-12 scripts cover their prior modules.

3 Deployment Documentation

The guide provides migration ordering, private database configuration, poster storage setup, lab account creation and an optional local Apache virtual host.

4 Security and Repository Review

The guide keeps credentials, database configuration and uploaded poster files out of Git. It calls for reviewing the diff before committing and avoids claiming a GitHub submission that has not been made.

5 Limitations of Current Verification

Public API and local smoke checks pass. Authenticated workflows need the lab administrator password. Live screenshots and accessibility review remain to be completed.

Implementation Evidence

The reports preserve the Practical 12 report structure. Practical 13 authenticated checks passed 9/9, Practical 14 admin dashboard checks passed, and Practical 15 smoke checks passed 8/8. No remote repository changes were submitted; live screenshots and student/faculty checks remain outstanding.

Database Results

Check	Current status
PHP source syntax	Changed PHP files checked with PHP 8.2 lint.
Fetch client syntax	JavaScript syntax checked with Node.
Local XAMPP deployment	Apache and MySQL reachable; app database connection verified.
GitHub submission	Existing repository identified; no push or commit made.

Testing Summary

Test input or action	Observed result
Changed PHP files	Syntax checks passed.
Fetch demo JavaScript	Node syntax check passed.
Practical 13 API integration	9/9 authenticated and public checks passed.
Practical 14 admin dashboard	Migration and admin checks passed; student/faculty menu checks pending their existing passwords.
Practical 15 smoke script	8/8 checks passed against the local site.
Accessibility review	Checklist and responsive checks documented; live review pending.

Screenshots of Portal Implementation

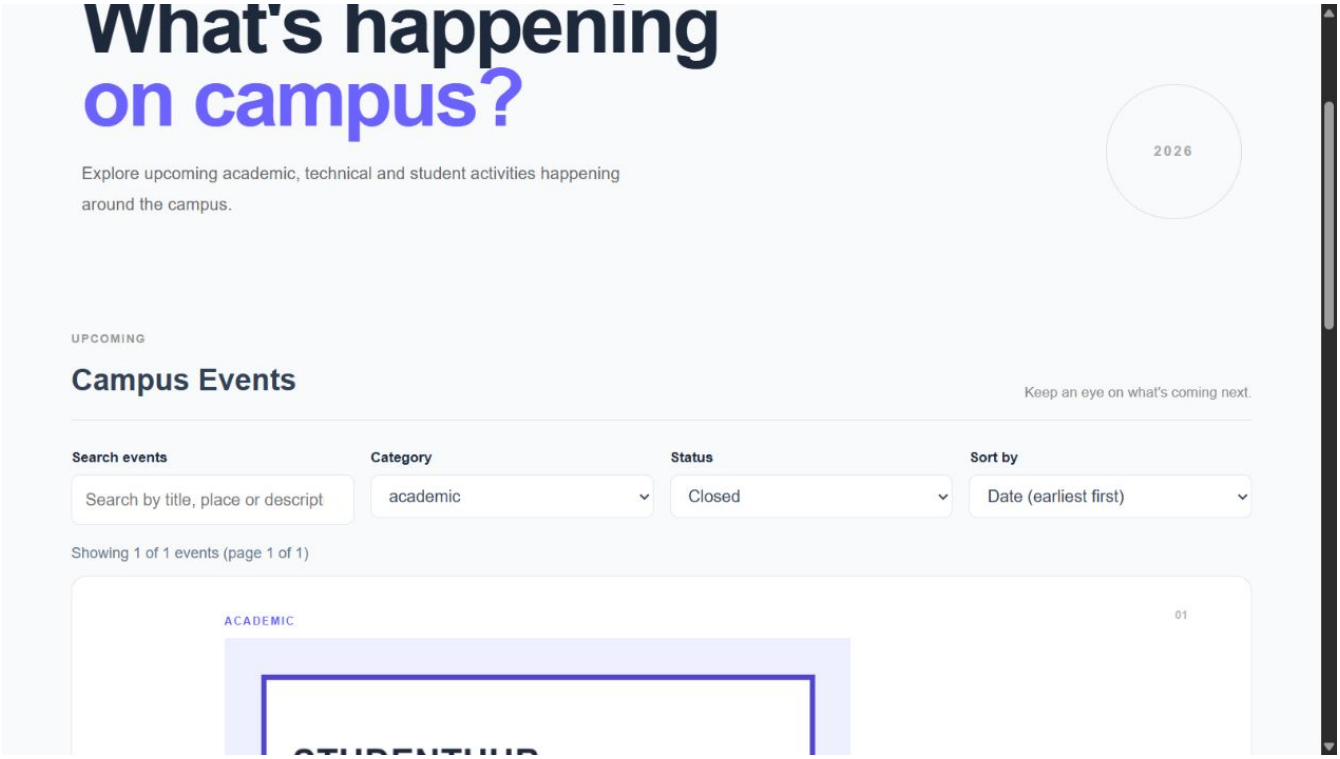


Figure 1: Existing public campus-events page used for the integrated portal.

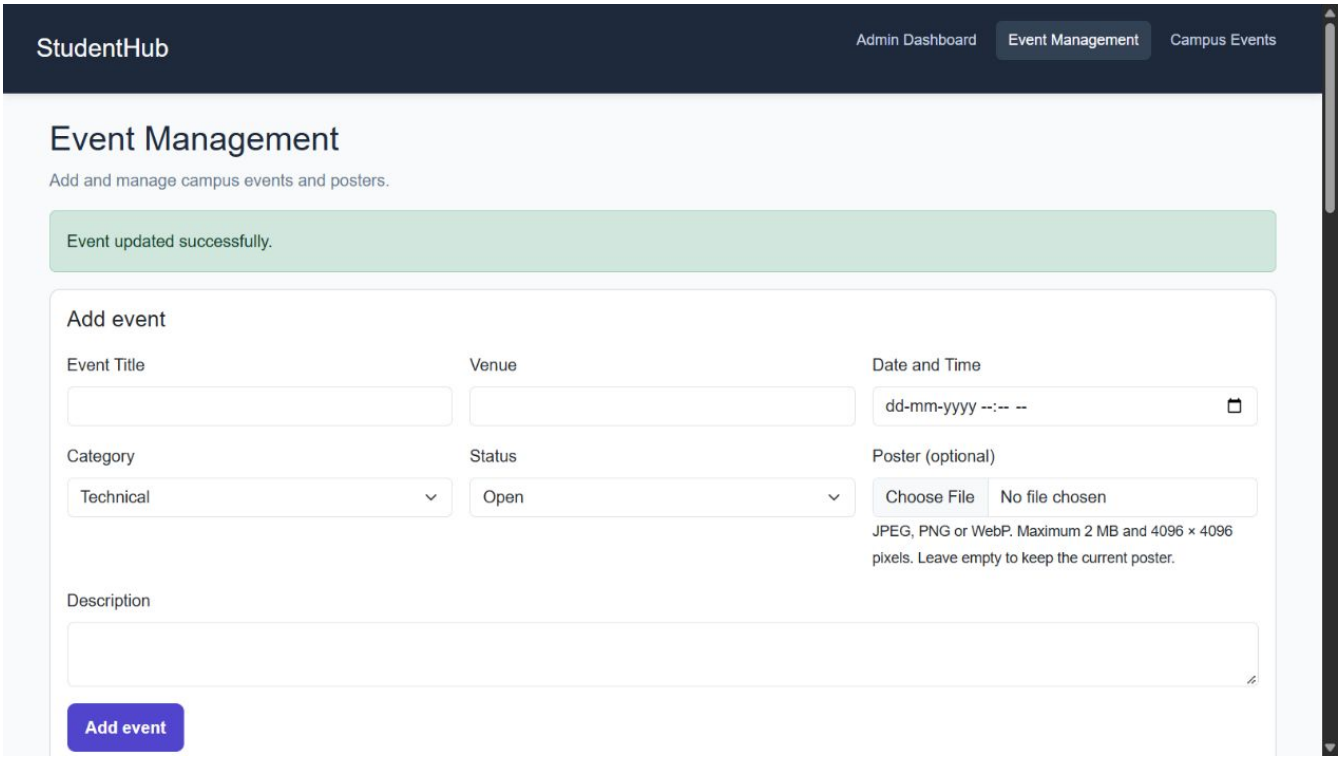


Figure 2: Existing event-management update state, showing the portal UI after an event change.

Key Questions and Answers

Q1 Are all pages properly linked and navigable

The new admin dashboard links to the student, event and API modules. The deployment guide includes a page and responsive review checklist. Live browser verification remains pending.

Q2 Are sessions, roles, validations and database operations integrated

Existing authentication, role checks and prepared-query CRUD remain in place. Practicals 13 and 14 connect their API and dashboard actions to the same database.

Q3 Are security and accessibility checklists completed

Security and accessibility review steps are documented. The interactive checklist cannot be marked complete without reviewing the running site.

Q4 Is project documentation sufficient for installation and testing

Practical 13, 14 and 15 setup guides describe the migration, API, local XAMPP and test steps.

Q5 Is the repository clean and committed

The existing repository URL is documented. No commit or remote submission was performed in this task.

GitHub Repository

https://github.com/DarshilRupani/studenthub-portal
Repository changes remain in the local working tree until reviewed and committed.

Conclusion

The local smoke suite and Practical 13 API suite pass, and the Practical 14 admin dashboard checks are verified. Student/faculty checks, live screenshots, accessibility review and final GitHub submission remain outstanding.